

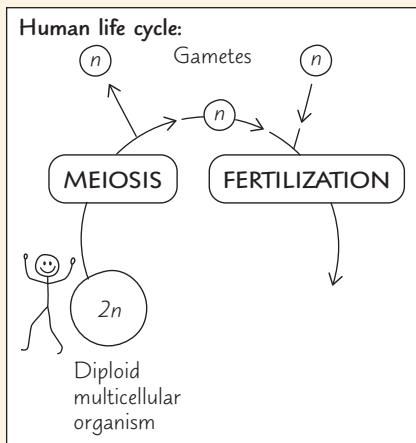
31 Fungi

KEY CONCEPTS

- 31.1** Fungi are heterotrophs that feed by absorption *p. 655*
- 31.2** Fungi produce spores through sexual or asexual life cycles *p. 657*
- 31.3** The ancestor of fungi was an aquatic, single-celled, flagellated protist *p. 659*
- 31.4** Fungi have radiated into a diverse set of lineages *p. 660*
- 31.5** Fungi play key roles in nutrient cycling, ecological interactions, and human welfare *p. 667*

Study Tip

Draw a diagram: To help you recognize key differences between the life cycles of fungi and humans, draw and label simple diagrams like the partial example shown here depicting the life stages at which meiosis, mitosis, gamete formation, a multicellular organism, and fertilization occur in humans and in fungi.



➔ Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 31
- Animation: Life Cycle of a Mushroom

For Instructors to Assign (in Item Library)

- Activity: Fungal Reproduction and Nutrition
- Scientific Skills Exercise: Synthesizing Information from Multiple Data Sets

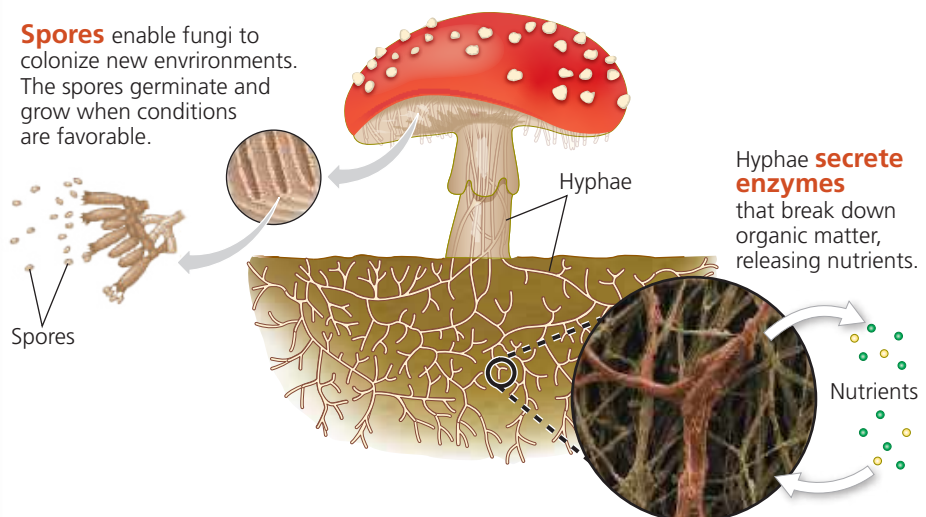


Figure 31.1 These little mushrooms are just the small aboveground extension of a vast network of filaments located below the forest floor. Such underground fungal networks, called mycelia, in some cases link mushrooms that are hundreds of meters apart. In fact, the largest known mycelium spreads beneath 965 hectares of forest—more than the area of 1,800 football fields.

How do structure and function in fungi relate to their role in ecosystems?

As they grow, multicellular fungi extend filaments called **hyphae** into their surroundings.

Spores enable fungi to colonize new environments. The spores germinate and grow when conditions are favorable.



Almost any organic molecule can be digested by at least some fungi, making them highly effective **decomposers in ecosystems**.